

Architectural Style Selection and Justification

Intercity Freight Load Matching Marketplace

Selected Architectural Style: Microservices Architecture

The Intercity Freight Load Matching Marketplace uses a Microservices Architecture. In this architecture, the system is divided into multiple independent services, where each service is responsible for a specific business function. The component diagram consists of User Management Service, Freight Load Management Service, Bidding and Matching Service, Shipment and Proof-of-Delivery Service, Payment and Escrow Service, and Database Service. These components communicate through defined interfaces and dependencies.

Reasons for Selecting Microservices Architecture

1. Separation of business functionalities:

The marketplace contains several distinct functions such as user registration and verification, freight-load posting, bidding, carrier selection, proof-of-delivery verification, and milestone payment. These functions can be separated into individual services. For example, the Bidding and Matching Service handles carrier bids and matching, while the Payment and Escrow Service handles milestone payments. This separation makes the different functions easier to manage independently.

2. Supports the complete freight workflow:

The system follows a sequence from posting a freight load to receiving bids, selecting a carrier, submitting proof of delivery, verifying the proof, and releasing the milestone payment. Separate services can handle these stages and communicate through interfaces such as the Load API, Carrier Assignment API, Payment Trigger API, and Payment Gateway Interface. This allows the components to interact while maintaining clear responsibilities.

Security Advantage

Microservices allow sensitive operations to be handled by dedicated services. The User Management Service can manage user registration, verification, authentication, and roles, while the Payment and Escrow Service can handle payment-related information separately. This supports controlled access to freight, carrier, shipper, and payment-related information. The requirements specify that unauthorized users should be denied access to protected information, making controlled service-level access important for the system.

Performance Benefit

The architecture supports efficient processing by separating different workloads into individual services. For example, bidding and freight-load operations can be handled by their

respective services without placing all functionality in a single component. The Payment and Escrow Service can also process milestone payment requests after the Shipment and Proof-of-Delivery Service verifies the e-POD. This supports the requirement that milestone payments should be disbursed automatically within 60 seconds of verified proof of delivery.

Conclusion

Therefore, Microservices Architecture is selected for the Intercity Freight Load Matching Marketplace because it separates the major business functions into dedicated services, provides clear communication through interfaces, supports controlled access to sensitive information, and allows different system operations to be processed independently. This architecture aligns with the marketplace's functional requirements and the performance and security requirements defined for the system.